ActInf Livestream #014.1 ~ The Math is not the Territory: Navigating the Fre

Livestream | The Math is not the Territory: Navigating the Free Energy Principle | 2:04:36

hello everyone
welcome to the active inference live
stream it is active inference live
stream 14.1
on january 19 2021
welcome to the active inference lab
everyone we are an experiment in
online team communication learning and
practice related to
active inference i'm going to share my
screen for
participants you can find us at our
website
on twitter gmail youtube or
our public key based team or shared
username
this is a recorded and an archived live
stream so please provide us with
feedback so that we can improve on our
work
and whether in the live chat or the
comments just feel free to
share whatever you're thinking all
backgrounds and perspectives are welcome
here and as far as video etiquette for
live streams
we'll mute if there's noise in our
background raise our hands so that we
can hear from everybody who wants to
share
and use respectful speech behavior so
for the active inference
live streams in 2021 we're going to be
having these regular
tuesday 7 a.m pacific times and then

we're also going to be having these
special sessions like the model stream
and several other kinds of events that
we're really excited to be planning
and go to this big red link and you'll
find a spreadsheet
and in that spreadsheet you'll find the
details on past
and present and upcoming active
inference live streams
here we are in 14.1 discussing the math
is not the territory with author mel
andrews
next week we're going to be discussing
the same paper for a follow-up paper
and then we're moving on to a different
paper but for this week and for next
week we're going to be in
this awesome paper and this awesome
group of discussions
today in active stream 14 we're going to
have
introductions and warm-ups then in the
sections of 14.1
we're going to be going through the
paper the math is not the territory
navigating the free energy principle by
mel andrews in october
and we have some slides with aims and
claims
abstract roadmap etc but this is pretty
much going to be whatever people want to
discuss today
and whatever questions we want to
address in next week as well
so if you have any questions just save
them and submit them
however you can and get in touch if you

want to join
live okay awesome so
let's start with the introductions and
we'll just give a short introduction or
check in and then pass to somebody who
hasn't spoken yet
and especially if it's your first time
we would love to hear
what your background or interest in
active inference is so i'm daniel
i'm a postdoctoral researcher in
california
and i will pass first to mel
hi there i'm mel andrews and
uh i think for better for worse my uh
doctoral work is going to end up being
largely about the free energy principle
and
active inference so
that's why i'm here cool pass it to
somebody who
can then introduce them
hi there thank you mel um hi my name is
marco uh i'm from holland
my background's in psychology and
concrete neuroscience
and german masters also worked a bit on
its independence
and my interest in active inference is
um
well a multitude of things i just i just
believe that it really has a potential
uh as alluded to and written about by
male
uh in the paper which doesn't today uh
in general just a lot of potential to
understand a lot of aspects of the world
in society that

we sovereign haven't been able to really
grasp with alternative minds
so happy to be here
then passed an opacity to yvonne
hi my name is juan i'm from moscow
russia
i'm a researcher in system
school i'm
i'm an engineer systems engineer and i
pass it to
hey guys i'm shannon
and i'm working in
i do brain stimulation and i'm also
interested in
social cognition and how we interact um
like in a musical ensemble
and how the free energy principle
applies to
both scales of those questions
and um pass it to alex
alex we can't hear you distance x alex
um but you don't appear muted let's go
to alex oh
now we hear you go ahead yeah hi
everybody
i'm alex and i'm in moscow russia
and i'm a researcher in systems
management school
and trying to find a way to find
integration between
active inference and system engineering
approaches
and i bless it to alex
okay thanks um thanks alex i'm alex
um i'm interested in uh i'm a
philosopher i guess
uh interested in free energy principle
and active inference and

sort of fundamental questions about
mental representation
and lately i've been more interested in
applied work and
sort of um combining eventually maybe
combining
these approaches with other approaches
to machine learning and so active
inferences like a technology
but um i will pass it on to um
let's see uh dave
dave you muted so we have dave and then
we have also then the last remaining
tim stephen i'm retired from information
technology live in the philippines
my background is in cybernetics and uh
machine translation i'm teaching myself
neuropsychoanalysis
and i was so pleased to come to mel's
paper because
i've been trying to figure out what is
this thing after all it's like a
a metaphor among metaphors among
metaphors and maybe
uh you can tell us toward the end of the
presentation whether that has anything
to do with what you've
what you're saying in your paper and i
can't even see who else is not spoken
uh philippe philippe there we go
hi i'm philip uh i'm studying
hero tickle biology in prague and i'm
interested in
computational neuroscience and i
joined because i only read mel's paper
and i found it interesting but i don't
know anything about it
perfect um let's go to tim

i'm tim i'm in toronto canada it's my
first time
joining in the sessions
really eager to learn about this stuff
was really fascinating and my background
is in
science and engineering uh specifically
i guess electronic systems and software
and uh that kind of thing and i'll pass
to stephen
go for it stephen or fellow jitzer
okay and sarah have you gone
hi i'm uh sarah davis i'm um in berlin
and um right now i'm a master's student
in philosophy of science but before that
i was an artist and the through line i
guess that's interesting to me is um
the relationship between information
and materials and media so i'm
interested in like reservoir computing
and how how and when you can separate
those things and
you know what is life like all these
funny things that are circling around
that so active inference is just yet
another
useful thing that um i want to
understand better because it seems to
connect to a lot of other
disciplines
cool it'll be interesting to hear about
that soon and then i see
blue or yes
hi good morning i'm blue knight i'm an
independent research consultant based
out of new mexico
nice is there anyone who has not
introduced himself

otherwise we're just going to keep
rolling all right great so
let's go to our warm up questions and
i'll just put up the first two which are
what is something you're excited about
today
and what is something that you liked or
remembered about the paper
and so people can just raise their hands
and we'll go in order
and until i see a hand i think something
i liked or remembered about the paper
was that there was no figures
so from the point of view of usually
reading papers with figures that was
very notable but then also that there
was clarity
and structure despite figures which are
often used
as a uh not quite an aid to structure a
paper
but it's how we often think about in
empirical research like oh what is
figure one gonna show
but there was clarity in this article
without the usage of figures so marco
and then anyone else who wants to
continue
yeah i fully agree that that is um
uniquely clear
so um i've been kind of lacking behind
on on the literature but
for me mel's paper is the best paper so
far when it comes to
contextualizing french principle and
like the title implies
really mapping it out right
contextualizing in the big complicated

epistemic
landscapes that this all connects to um
so it's a huge props to know for that um
i think the only other critical piece
i've
read is from colombo which is more
unwieldy
um and i think uh the the kind of
importance of
mapping really shows in this paper
because it it opens up a lot of
directions
to think about the fdp which stands in
stark contrast to a lot of other
papers are read which kind of uh take it
as authoritative and then just
move on instead of uh actually
trying to interpret it and so mel does a
great job at guiding the reader
and understanding the nuances and the
questions that would arise on
when you see those nuances so excited to
hear about
people's thoughts today cool very nice
uh
tim and then anyone else who raises
their hand
i thought that i was i i noted to myself
just you know
as i was reading how clear the writing
was i thought the writing was really
clear
and and expressive and explication
really well ordered
and um and there was kind of like a i
guess uh
a humility i remember somewhere in this
content i saw the idea of the list of

epistemic values i think
it was you call them and humility was
one in the list and that kind of comes
through
in that paper the way that uh there's no
sort of diving for like a
discovery or a conclusion it's really
just laying it out and
and and giving it to you to to
leaving the questions open which was
really nice
yep and also lots of nice philosophy
type things like the idea of defending
an idea and so
it's not common to see in an empirical
paper like we're going to defend this
it's it's a lot more like a battlefield
of ideas in philosophy
and in science it's a little bit of a
different structure so sarah and then
anyone else
to like lather on the praise of like
uh to mel's paper but yeah the writing
was really good and actually when i
first
read it um i i feel like i understand a
lot more
than when i first read it and so now i
kind of feel like i'm happy to be here
because i need to read a second time
i've learned a lot more
um but yeah the x the exploration of
kind of the beginning of the exploration
of all these different models
it was um uh saw a lot of potential
cool and that's kind of part of the idea
with a dot one and a dot two
is sometimes you read the paper and you

don't talk to someone about it you have
thoughts one two three and then you have
a first conversation about it or you
listen to one video and all of a sudden
you have
thoughts one through ten and then you
have the discussion and all of a sudden
you're
you know up to 20. so that's part of the
idea of taking the time
to really read through these papers um
does anyone else want to add anything
i'll throw up the third
warm-up question which is what is
something you're wondering about
or would like to have resolved by the
end of today's discussion
so alex kiefer then anyone else
yeah so i think having this resolve by
the end of today is probably a tall
order but
i'm i'm interested in the question that
sort of
um explicitly raised it toward the end
of the paper about whether the fep
is a very abstract sort of
uh model of or in some sense maybe
representation of
nature or a pure formalism and like
that's an interesting line
so i think it's an interesting question
great question
sarah then marco
yeah i was i to me it all just seems
like mel fix it by the end of the hour
but um but i yeah i was curious to know
like if
if you have any insight into ways that

the model could never be applied you
know like just
trying to situate the model in all these
worlds that it
has potential for
cool any other intro thoughts there
uh marco and then or anyone else
yeah um a lot of questions for mel i
think but but i think what maybe
i hope so uh possible to to resolve by
the end of this stream is
um there's only a very brief note about
it but but you also said that it's also
compatible with processionism something
about possessions i mean they
i didn't see anything else about it so
i'm not a philosopher so natural kinds
for me always
was a bit a weird iffy concept for me um
but i was wondering if you could resolve
maybe um
the problems of mapping it to natural
kinds of stuff but
for professionalism because it seems
that activation is much more
sexualist kind of framework not really
compatible with the
essentials objects properties framework
of
ontology um so yeah i'll leave it at
that
cool let's definitely return to that and
define it and
get clarity on it shannon than anyone
else
hey yeah so i'm really interested in
what role the free energy principle or
active inference

plays in empirical
hypothesis testing um and
i think this paper did a really good job
of explaining that the mathematical
models can be
very useful to generate
questions to generate hypotheses or
just to further investigate some
empirical phenomenon
and i think it would be good this week
and next week to just sort of talk about
what it means when you say i have a free
energy principle characterization of
system very nice
agreed and um yes we have several
fun hours of discussion ahead so i'm
just going to
flip to this summary slide and ask
mel when you're conveying this work to
different audiences how do you
communicate it or what would you like to
start with just how do you give your 30
000 foot
overview perhaps to different audiences
or perspective
yourself in the past yourself in the
future however you want to do it
yeah i guess i saw a lot of um
confusion about what the fep was and
what it was doing um and it was
my before i could really uh
play with it i had to um i had to get
clear
on what it was and and make everyone
else clear on what it was
um and and so i
i tried to tackle kind of like the most
fundamental questions in this paper

about what it is that the ftp is and
what it's doing
um and in some ways
in some ways i think a lot of those
remain
kind of unanswered by this paper
but but no longer unexplored like i i i
i put it all out on paper so you know so
we can talk about it now whereas before
there are these things you know like is
the fep
science science like is this a
scientific model or scientific theory
or is it you know that the demarcus like
does it does it not belong to the demand
of science
um that that was a big one and no one
had asked that
um and then uh
well alex's question that's what you
know and and i didn't come
i i i sort of write the first sentence
right
is saying i think it's the main explains
but then i'm gonna i'm i'm
i'm writing a paper with an as a photo
um
where we're gonna come down kind of
differently on that
on the i mean you know it's not written
it's not 100 written yet we're not true
but but we come
down a little differently on the
demonstration problem and then um
uh in terms of alex's question
that was that was one of my main driving
questions the notion of
as if is the fpp like a really abstract

representation of natural systems
central processes
or is it not a representation of
anything in nature at all
and that to me is like totally not
answered yet
um and there's some there's almost like
a queen
like that that's almost to me it's like
getting to the heart of like
naturalism as a as an approach in
philosophy like
like uh carl's carl friston's line
this whole time has been well there's a
dissolving line
between just math and
and physics proper um
and that drives me crazy but uh
but i'm coming around to a little bit
that wasn't a good summary of my paper
that was a good summary of
of what i'm excited about exploring
further
fun yep well just a few things in there
that
might fly above or below those who have
different
um you know listening regimes of
attention
you said before i could play with it i
had to be clear on it and also help
other people be clear
so that just captured this learn by
doing learn by playing
and also learn by serving and teaching
because those are some of the key
pieces and without being playful and
understanding that it's a deep well

to dive into and then also that there's
a community that is also curious those
are some of the key
aspects and then also it almost goes
without saying that this is a really
exciting philosophy paper and it's an
ongoing philosophical question
so people are maybe familiar with a
measurement is made on a protein or a
new species is found in some forest
and it gets nature science and people
are really excited about a measurement
that's made about nature
and then the philosophical questions
potentially because they're fundamental
or there's no simple sound bite answer
or there's no figure there's no
graphical abstract you got to read the
prose or got to think through it
the philosophical questions are often
seen as not contemporary
or not exciting or not cutting edge but
we're seeing
in real time with free energy principle
and active inference
the way that maybe we could even call
them top-down philosophical priors
on what the fep is are shaping
the realization of what emerges from the
bottom up
for example if it's really from on high
and fep is part of science and it's part
of a scientific
program that's going to be very
different several years down the line
than if people said this should be
thrown over there with the fiction books
and

you know with the biography of somebody
else so it's pretty interesting just how
these
top-down and bottom-up ideas are playing
out
in the discussion about what this
framework is
so pretty interesting and anyone can
raise their hand or give a thought or a
question the other
piece i just wrote down until anyone
raises their hand was this
unanswered but not unexplored which is
taking us back to the map metaphor and
it's the math is not the territory of
course
a little bit of a pun on the map is not
the territory
and what is the map what is the
territory
and where are their dragons where do we
have just the coastline but not know
what's on the interior
or where are the underlying features of
the world
stable or changing and then where's our
map stable or changing
in an attempt to understand and just
like the navigation there's probably
pure navigators out there but also
there's those who want to
move cargo so how are we going to get it
done with this map
and this territory so we'll go to alex
kiefer
sarah then mel
um yeah so i guess i'm just following up
from some of what mel just said um

um so so i don't want to take this too
far in the direction i'm thinking about
because it's it's pretty
general so so i to me this this paper
raises questions that are
uh fundamental like whether or not
you're interested in the fep
and then it's it's partly a matter of
seeing how these two things fit together
but like
so mel mentioned quine so one problem
that i've always had with the
with the line that the fep is not
falsifiable or that it's like
distinct and kind from active inference
as a process theory is just
like i guess confirmation holism so
uh so i just assume that um if this fep
thing is informing our empirical
theories at all
then there must be some semantic logical
connections of some kind between
the empirical theories and the fep so um
and if
you're if you're a confirmation holist
do you think that you know
sensory evidence uh what we observe kind
of
back propagates if you will to like all
of the beliefs in our network
then i think the fep should be
vulnerable to that but again that's not
really about the fep
so um i just want to sort of raise that
issue more explicitly and
um i don't know cool well just before we
go to sarah could you just define a
little bit

better what is the confirmation holism
oh yeah
who who would think differently on this
issue and what would be the implications
for active inference
well so i don't i don't know what people
think about this issue these days um
actually i'm just this is sort of like a
i guess it's attributed to quine and
do do him i'm sorry if i'm
mispronouncing his name um but the idea
is just that you
that sort of scientific theory um does
it like it meets sensory evidence as a
corporate body so like
the idea is that if you observe
something that's in conflict with your
total theory
there's no uh
i don't know way a priori of telling
which of the claims in your theory you
need to revise
right so like if there's i mean you
could
you could for example you could say well
there must have been an error there must
have been a malfunction in the measuring
device
and so you could you could revise that
part of your theory or you could say no
this really does
falsify my more fundamental assumptions
and so
that's kind of what i'm getting at and
i'm not sure uh if people hold
of the alternative view these days that
like no there are certain
statements that are meaningful but

immune to falsification
but that's the kind of question i'm
raising excellent uh
sarah mel marco
i think actually just to not overwhelm
mel my
question could fit in anywhere i'd
rather just give mel a chance to
yep let's let's go back to mel then
sarah
yeah the the web of belief thing yeah
the the do him applying hypothesis is
this idea that that we've got an
integrated web of beliefs and you can't
test any atomic proposition in isolation
from you can't you can't protect
anything in your web of beliefs
from from revision with by by new
evidence
which i think it's almost like very uh
compatible with
with this kind of asian frame view right
there
you get this you get this spreading its
propagation
of of evidence through the network all
the way up and down
um i like that idea
but um what did i want to
respond to oh yeah someone said
someone said active interest is a
process theory and i just want to like
set the record straight on that um uh
the the reviewer one on my magazine i
just i just got reviews back on this
um it's not published yet i just got
reviews back
and they pointed out that um it it's

probably misleading to refer to accident
prince of the process theory
there have been process theories
on the basis of active inference and
variational message passing
developed so when you see the like the
there's a paper fristen schwab uh
rivoli joe
sulo is on their um 2017 it's called
active influence across the street
that's that is a process theory that is
made from active inference that's
that's set in um variational message
passing
but active inference itself is not a
process through so much as the corollary
of the fep
applies to to to activity in an
environment
and then processes of that have been
made so i
i should probably i'm going to go back
in and like be more
careful with the words i use
cool marco then i'll go ahead
yeah um i'm not sure if you met me with
their process theory but but i i i
intentionally also avoided the phrase
process theory because i think it's
what i was trying to say with that
comment earlier is
a lot of the treatments of natural kinds
and mapping to real states or affairs of
the world
seems to me more grounded in a more
object property oriented
philosophy pathology right but for me
it sounds more that's the the

ontological
uh assumptions or commitments in
processionalism are far more
compatible with active inference which i
wanted to address it later i'm not sure
now it's very time but but um
uh there's this issue of not demarcation
for
is fap ai scientific but also
demarcation of markov blankets right so
as you pointed out rightly
there's this dilemma of you know where
do you put the threshold for
where the boundary of the marked blanket
is but for me that fuzziness
is not an issue if you would be more
professionalist because then the claim
with the necessity would be more there
is a boundary
but that we cannot affirm or confirm
where it exactly is
what the you know perfect absolute
stressful is it's not an issue itself
simply that there is heterogeneity there
is certain
boundarying of states would be
sufficient i think
because the procession or dynamic nature
of these systems
uh would mean that you can't say there's
a static on simply that there are
boundaries between
subsystems but but that was not my
question my
comments um for now was basically one
thing i i
think is also interesting is um seeing
fpai as simply a language

so there's the canonical kind of typical
notion of math is a language a physics
it's a discipline in which
that's written in in maths and i i kind
of
feel that this is the same here so
daniel gives very beautiful metaphors of
navigation and mapping
and to me it seems that uh this whole
project
and not just theory but the whole
project of p and active inference
is more of a vehicle of toolkits of
languages and allows you to translate
back and forth in disciplines
translate into mathematical abstraction
and as you explore in the mathematical
abstract world
you can translate back so it's like if
you can't see the patterns in empirical
data
translate back to abstraction find
patterns there
um see if you can translate them back
and then you can test them and so it's
like the
scientific narrative with a detour but
um yeah i don't know if you find it
interesting would love to discuss that
more later
but yeah yeah that it is interesting and
uh also the navigation brings us back to
cybernetics and the the word cybernetics
comes from
navigation and so it's almost like if
you can't navigate on the water
surface and then you go you abstract
beyond

some sort of limitation and so it's kind
of funny because the mapping metaphor
again
it is a physical one it's related to
cybernetics but then also there's this
idea of a map
in mathematics and a mapping is a
function and so there's this abstraction
abstraction mapping and then there's
also
that as a map onto the physical world
because we're embodied agents
so if mel or anyone else wants to
raise their hand otherwise i think
there's a few other threads to highlight
there go ahead mel
yeah i just want to flag something marco
said quickly um
the markup blanket isn't actually all
that it's not actually
as a boundary the markup link is not
actually all that diffuse and dynamical
it's actually
it's actually harnessed to like
particular
particles and you know in a sim so
we think when we think about you know
the boundaries of an organism
or the boundaries of a cell wall or
something right
you're constantly you are dispersing the
cell is dispersing right
the material of which the
cell the organ the tissue the human
being
the society is made up right these are
these are
these are diffuse and they are

dissipating they're the material there's
a constant turnover in
like the material composition of these
things right
um and and the markov blanket doesn't
handle that right now right the markov
blanket is ropes to particular
particles so to speak in
in your system so it's um
there there's a there's a a kind of a
desire
in um in the original like burkhoff
stuff on dynamical systems um you get
you get uh 1927 first off 1927 you get
um
he kind of proposes a notion of
wandering sets
and um there's kind of a desire
carl has a desire to extend the markup
blanket
so as it so to be able to accommodate
wandering sets
where where there is a turn right that
where the
the material makeup goes away and yet
the boundary persists
right you get turnover but um it doesn't
do that yet so it's actually
it's actually it's not nearly as like
dynamical as as you'd want it to be i
think
the the wandering sets and the
attracting sets it kind of reminds me
again with a navigation metaphor
the attracting set could be like there's
like a whirlpool and all the boats
around it are kind of being drawn into
this attracting

set if they move away then they're drawn
back in and then
the wandering set is like a flotilla of
ships maybe some are
joining and some are leaving but they're
maintaining a coherence
as they dynamically are moving through
the ocean
or as their subunits are changing
because there's this dissipation
that was brought up and then the living
systems that are far from equilibrium
are the ones that are apparently
successfully
resisting that dissolution and so what
is it
that those living systems are doing and
that's the what is life
question that's schrodinger 1944
to maxwell 2018 that pipeline so it's
it's really
interesting questions marco then any
other raised hands
thanks uh and um thanks for the
clarification um maybe i'm
misunderstanding something but
um so so i guess for me maybe it's my my
personal
interpretation of it all but for me the
primacy is actually less about
the formalism as for example the pro and
originally uh
formulated as but what for me is
interesting is more
the conceptual interpretation of there's
an intermediating
kind of interface right there's a
unidirectional flow

um that's that's in its dynamics
allow for this protection of states that
shouldn't
go beyond certain uh boundary for
example
um and in that sense it oh wait and one
more thing is that
maybe that which is to be bounded or
protected
doesn't have to be a particular state as
such maybe it's like a relation
for example you're protecting the
property of certain nodes of certain
states or certain densities
that they are adaptively viable for
other systems that they might be in
relation with
um but but i think in general i think
what this all shows is that
there's a lot of freedom at least in my
opinion to interpret the picture that
follows from ffp ai
i'm still kind of unclear why it's not
dynamic
because the mathematical nodes can be
variably mapped
to physical nodes in my opinion because
i'm not sure what you mean with
particles right so
it's something like we we do the
boundary centers of particles but more
in terms of
sensory states or or or that through
which
um the activation inference is mediated
um but yeah that's my but i would love
to hear more about
you brothers cool if anyone else

raises their hand i would just kind of
almost take stock there's a few
threads happening here there's some
philosophy of science
questions what is a principle what is a
theory what is falsification
all these types of questions were um
brought up from the more philosophical
side of the literature and then there's
this
a little bit related discussion about
markov blankets dynamical systems the
definition
of empirical measurements and how
cleanly we sort them into
these different types of uh natural
kinds if we want to use the
philosophical
word or maybe just a functional whole to
be less
essentialist about it there's a few
different threads
happening here so it's just good to keep
track of these
pieces and uh yeah if uh steven i saw
your
hand raised or anyone else yeah hi
thanks
um i suppose one thing that i've been
quite interested in
is dimensionality in the sense that
entropy is a dimensionless sort of
property
so you get this interesting question
that so the free energy principle
in a way is mostly dimensionless or
it's like some sort of trend and then
when

that comes into active inference it
doesn't get fully dimensionalized
well it doesn't it's like lots of
different active influence processes
each one can contribute to
dimensionalities like multi sort of
threads
the actual signal unlike when you
mention like say in cybernetics where
you have something that is guiding you
the signal itself doesn't have
you know as we say the signal doesn't
contain per se the data
particularly the entropic noise it's
through inference
of the dynamics within
the noise that the generative model can
sort of
then process that and so it sort of
raises this question about
you know when things go from one level
to another
blankets in these states um
is that transference between states also
dimensionless
in that sense or you know where did
dimensions start to come in as we go up
the blankets or
maybe as things get aggregated
interesting question and that reminds me
of some other questions that have been
raised about the dimensionality
of active inference like is each sensory
modality
enshrouded in its own blanket or is it
sort of a tuple that gets passed across
a single
organismal blanket that represents

multiple sensory
possibilities so interesting
questions anyone else want to raise
their hand or we can just
start to slowly walk through the paper
and just
any thoughts that people have will raise
them
so we checked out the abstract
in the dot zero video here's the road
map
so this is a pretty thorough road map
because we can see
a lot of the sections and a lot of the
terms that come into play
so mel i have a question what
was your intention with writing it you
said that you wanted to clarify it and
learn as well
how did that translate into this
structure
and then maybe where are you headed you
know
off this road map this was one journey
what is your next
journey yeah
so a friend brought my attention to um
to the free energy principle and i think
fall of
2017 which would have been like my
penultimate
semester in undergrad and i was taking a
course on like mathematical psychology
so
um uh if if you
one of my references is right so the
reference to this normative model
this normative versus descriptive

process versus
right that that that reference is argon
can
minus 95 which is um he's he's one of
the primary figures in in maths like
so a lot of cognitive science you know
has had subsumed
math like but but the idea of like
having these formal mathematical models
of um
of cognition so that's that's what i was
immersed in so so i
i was uh that's what i was thinking
about most at that time so like oh a new
you know
a new mathematical model of psychology
perfect um
and i i i was very skeptical at first
and i dove in
and i um it
it brought me under it was really
there's a lot there
and um and i really struggled
so at first i really wanted to do that
too like you know it was immediately
apparent to me that there's this big
complicated thing that
that very few people understand and
there's a lot of widespread
misunderstanding of what it is and what
it's doing so i tried to do that at
first and i came up with like a little
uh a little like dossier on on like the
basics of the ftp
in like some of its early formulations
as an undergrad
um and i put that up and
but it it didn't solve kind of a lot of

the bigger
uh i guess like philosophy of science
questions about what it is and what it's
doing
um so i this is like a second stab
at that and and from here
um i i so i was thinking of the fpp
one one way it's helped me to think
about it is is as a model
and i don't i
i guess i don't view um
models as strict
natural kinds or like strips the
scientific kinds
i don't i don't think that there's like
a a list of necessarily sufficient
conditions for being a model or like an
essence
to being a model and the fep is either
you know
definitively a model or not i think i
think it's helped me quite a lot
to think of the ftp as a model it's it's
uh
reduced my prediction error if you will
so think of it as a as a very abstract
formal generic target list model um
but but you know and i think in a lot of
ways thinking of it as a modeling
language or a modeling framework
is also really fruitful i think that's
important there's not
there's not a philosophy of science
literature on modeling frameworks or
modeling languages right
um but but i think that's it it's
probably more of a modeling framework or
modeling language than it is

a model per se um and and next so i i'm
i'm next i'm going to try to work on a
paper with um
within us kind of exploring what it is
to be a principal
right like what's the principle because
there there isn't a philosophy
to run on principles really right what
is the principle
um that's next i also
um there's been a lot of comparison of
the fpp
to to natural selection to darwin's
theory of natural selection
and um when you ask people about that
and this is this is sort of the the the
core
like pro fep the core like
we're working on the fpp camp um
often draws that comparison and i think
that this
so so what they mean to be doing there
is saying well it's not directly
falsifiable
to alex's point i think i think
everything is sort of falsifiable every
everything in science is falsifiable in
so much as it's
if it if it doesn't get traction it
falls out of use if it's not useful
then we don't use it right and then it
it
ceases to be kind of part of our
collective consciousness as
as researchers and i think that if if
the fep fails to be useful
you know it falls by the wayside that's
that's

gets confined to dustbins of history um
so so there is that in that sense i
think there definitely is that kind of
like
confirmation holism there right um
if it fails to produce useful lower
order theories than it
it it's all that of use um
so so i think that that in in trying to
draw a comparison between
natural selection and free energy
principle people want to say
well the fep like natural selection is
not
directly falsifiable
um in the way that natural selection
isn't directly falsifiable
and it's um it's also
supremely useful um it has this
widespread kind of both empirical use
and
and you know grand explanatory power
and i think that this is sort of
ultimately conflating multiple
notions of natural segments so multiple
kind of discrete formulations of natural
selection
one of these due to popper and one of
these
one of them sort of due to to darwin and
then it gets formulated in like lewinton
and got christmas and stuff like that um
and one of them that safely do the
proper
which popular later kind of recounts as
a as a mistake it's an error news i'm
thinking
um so i want to do that and i i also

think there's kind of a metamark
because i've been working on models and
um
reification in models so so when we
what's happening with the ftp we can
think of as reification conceptual
reflication or pernicious relocation
where
you get uh confusing the map
for the territory basically perfect
thanks for that response and also
welcome uh scott
we will go to alex kiefer then marco
then tim
okay um so i guess
there are a number of things that i want
to say um so
as far as okay right so i think what mel
just brought up
um reminded me that there's a
there's a um one way of of i guess i've
discussed this with mel and others so
i'm
um apologies if this isn't in the paper
i'm not quite sure i need to look at it
again but basically one way in which
this thing
could fail to be falsifiable is if it's
not so much a claim as like a
an approach or like a a sort of
stance that leads you to make certain
claims so like
um so to put that in a if you wanted to
put a negative light on that you could
say it's a moving target but i
i don't mean to disparage the fep at all
i think i think
um you know one thing that's going on

here is that carl fristin is a fairly
creative
scientist obviously he's sort of trying
to do something new and so he's
you know this this is a framework uh mel
mentioned
thinking of this as a modeling language
um it's a framework that's still under
construction to some extent and um uh so
you could say it's well
it's not falsifiable because it's not a
particular you know mathematical
description it's a
uh generator of descriptions like that
um
so i'll stop there for now yep and just
to draw one
quick thought there it's the principle
as being metaphor
uh the metaphor is between the principle
and the language so could anyone
falsify english could anyone falsify the
idea of english grammar
or is english a moving target is it
something that can't be falsified
but you can make claims under the
umbrella or using the english grammar
approach
that the claim could be useful or not or
it could be useful for one person or not
for another person
so once you get into the specific
process theories the specific claims
then you're talking empirical data or
you're talking about the perspective of
individual researchers
but then when you pull back to the
principal level and mel it sounds really

interesting about the work on fleshing
out what a principle is
it's a little bit more like you can't
falsify python but there's programs that
don't work
okay so marco tim and then dave
thanks um so first of all apologies in
the fans with philosophers because i'm
gonna consciously tread into
pragmatism um so so again i'm not a
philosopher but but i'm
kind of interested in in the angle of
pragmatism on these matters
so um reiterating again the theme of of
seeing
the fringe principle and active
ingredients as more of a language
and a framework um we
also i think obtain an interesting maybe
mode of falsification that actually mel
just
gave us which is um if certain approach
or framework or
a system of doing science um is selected
by the niche that is a scientific
enterprise then that is maybe an
interesting form
uh well the opposite of falsification
right so if it doesn't
get adopted then that maybe is some kind
of classification
but then an interesting uh implication
here is
following the popularity and the strange
like popularity you must have
uh of sap in so many different circles
um is i think a sign that
that the scientists or the researchers

who have utilized it have
experienced an embodied signal of
usefulness
and then following that comes a question
of okay then where does it come from
right and then for me what's really
really interesting is then then the fap
is the perfect kind of approach to
answering or approaching that question
right so so upon adopting or upon being
exposed internalizing the ftp and active
influence
what changes within so now we get also
two kinds of approaches to language not
just you can communicate to other people
but also how do the different models you
you
you embody communicate with each other
so the unifying
uh capacity of free energy principle
is not just uh in the abstraction in the
papers
but also for people themselves at least
for me and anecdotally also others
that is i think one of the more unique
aspects of active influence it really
evokes the sense of unification
it allows you to use a single framework
and lens
for a variety of phenomena and theories
and to me that is maybe the biggest
argument for it being scientific for the
merits it evokes are genders
in scientific enterprises including
their constituents such as people and
resurgence
um so uh yeah so
i would love to hear more from you uh

especially philosophers on pragmatist
perspectives sorry i've been binging a
bit of hasso
chan recently so maybe that's why i'm
asking thank you
marco tim then dave than anyone else who
raises their hand
uh did you say pragmatist just now
yes that was pragmatism yeah a good
segue to what i
wanted to mention i guess because i was
when you mentioned rarification
that resonated with thoughts as i read
the paper
about um whitehead's fallacy of
misplaced concreteness
you know the idea that this this this
knot
in the processes that exists that's
being revealed
um is actually a process and the things
that we see in the objects of the world
are really just um uh sort of a
rarification of the processes
and you mentioned natural selection that
was another thing that came up but not
in the way i don't think you were
mentioning it now
the way i thought of it in the reading
the paper was uh basically the
objects as we call them are simply
uh processes that we can take a
perspective on that survived
you know they're there because they
survived they stayed they became
they sustained themselves as you like to
call it you know self-organized and all
that right

um yeah so anyway interesting thanks
yeah thanks for
tying that dave and then alex kiefer
yeah mel uh you don't seem to give the
exact
title of the 1995 loose
paper what was that i've got his 95
book but he doesn't seem to ha is it
just the introduction to that
uh um the tarot window
collection oh mel you're muted but
look it up and then post it in the
youtube chat or comments not the jitsie
chat because it will make a noise so
post it on the youtube and then anybody
who wants to post
any of these papers they're mentioning
really the place to do it is the youtube
comments
or send it to me and i'll put it in the
youtube video description
so that everybody can access it alex
kiefer then anyone else who raises their
hand
uh yeah i guess i was just gonna create
some more problems for
the simple discussion here um
and well in that okay so so pragmatism's
on the table and ratification
um so um i guess for similar quantum
type reasons i'm not sure how to think
about these things
i think what tim brought up about um
perception
forgive me if i could get it wrong but
perception being something like a
ratification of an ongoing process
i think is a really good point so like

um
i mean there's a sense in which i'm a
fan of ratification because it's like
yeah maybe that's just what we're doing
when we you know we take some we have a
language
uh maybe it's an internal neural
language or something or maybe it's an
external one and we and we take it to be
about the world and that's our lens on
the world so without rarification in
some sense
i don't know um how we we would have
any any kind of perception of things but
i'm sure that there's a way to
distinguish bad kinds of ratification
for good kinds
um and i was going to say something else
but it's i completely forget so
yeah good point it's kind of like
stability and plasticity
remember those two top level descriptors
of the cybernetic paper that we heard
from in the last couple of weeks
and it's kind of like they're not
intrinsically good or bad because you
could be too plastic or too stable
and succeed or not and then beyond
whether something exists or not humans
have this kind of second layer where
it's like oh yeah that exists but i
don't like it
so it's perpetuating itself but then we
just don't prefer it
and so that's just our agent's level
perspective and so it's not just enough
to be
reified there's something else happening

there so marco then sarah
yeah oh to build on uh alex's great
point
so one of my favorite notions is
reconstantine ladders
or in buddhism they also call uh talk
about the simile of the rounds
um and i think this is really what it's
about right so we're kind of like i
like i said we're kind of stuck with
reification that's just part and parcel
how we engage with the world because the
alternative was believing in nothing and
then
if you literally believe nothing then
how the hell are you going to engage
with the world right
so all you can do is have good re as
good as possible reification or
at least worse reifications um
and again for me this is this
perceptually this kind of approach
or at least perspective that we are just
engaging in finding better
reunifications
such that that particular reification or
such that that particular
um embodiment or incorporation of a
model leads to better genetic dynamics
and then to better active inference um
due to use active infrastructure
framework here um yeah sorry yeah
cool sarah then mel
um i don't know i is this possibly gonna
be an embarrassing moment for me but i
feel like that's my job
um i don't really understand
like as i was kind of stepping through

fep the math of it
there's a there's a point where i'm like
eh it seems like bayesianism has
got some like it doesn't seem like even
it
the foundations of what feps are a lot
of things are based on
i mean this idea of of belief and priors
and
and and that something you know
something is absolutely true
i guess i just wonder mel if you you
know if you
ran up against that or like if there's a
deeper background that i'm
probably not aware of in philosophy
about the limits of bayesianism
um or these kinds of models so that's i
don't it's
really a vague question but it's the
best i could do
no that's an excellent question oh sorry
yeah i'm nice it is it is a great
question go ahead
yeah it's an excellent question so
there's a um
yeah i think i think when i was first
grappling with veganism
there's a way in which um
there's sort of like a laplacian demon
smuggled in
like there's a there's a view from
nowhere smuggled in
there's a a a perfect knower
smuggled him and
that would be the case for the fep
but we've we've
we worked out a way

for it to be just a
a cognizer that's like
self-supervising right it's it's
knowledge is limited
it's access to to the true world out
there is
is intrinsically limited and yet it's
able to self-correct and self-supervise
and if you want um i think sort of the
best recent explanation of like
why this is the case and and how it's
able to do that is um
yuck hoey's paper and i want to say
synthesizer and i want to say
something self-supervision and the fish
i guess i have to i get i guess i have
to drop this link as well let me go do
that
cool yeah that's what's so fun is the
discussion yeah i would love it thank
you
yeah the discussion we raised so many
fun links and so many connections and
then we just
make sure to post them so that everybody
can asynchronously or synchronously be
in the game so marco then anyone else
oh no raised hand yeah oh go ahead uh
oh yeah sorry um what i'm gonna say oh
yeah and i really love that but i
i i wanna propose saying instead instead
of a perfect knower because laplacian
indeed knows everything but for me
bajanism is really more about
if we would have to make an analogy it's
a perfect pragmatic learner
so in essence what it it the nordic
principle kind of

by design says this is the perfect way
of knowing
if you also add on a lot of assumptions
for a particular age
but but uh yeah i guess a bit half
jokingly you could say it's a fristen
steaming
you know the perfect dramatic learner um
yeah i just
i just ran up against this just in my
everyday life
i contradict myself in so many ways i
can't even keep track you know like
one one part of me thinks this the other
part of things that and they're
literally
and so i yeah i just i just started
realizing
my my own limitations with respect to
knowledge and how knowledge would even
be modeled
but that's part of the design i think
that's something i often like to point
out
is it's not about having no conflict
it's more about given that conflicts
will arise
how to best cope with them so the
adaptive engagement with the world is
also reflexive
so your own struggles your own
challenges your own
seeming contradictions are also
opportunities to grow and that's again
only possible with this
demon that that is pragmatically
learning all the time
i'm curious but hopefully that's a bit

more optimistic yeah i'm curious about
this perfect demon
because i would say that a bayesian
updating agent
could make a update that helps it
survive in its niche or not
it could over learn or underlearn and
just simply fail to exist in the future
and so natural selection again is the
hand that just sweeps off the table
the bayesian learners that aren't
existing and i would challenge you know
there's many thoughts on bayesianism
and a lot of debates but alternative
being what frequentism
or some other vaguely unspecified
mathematical framework that somebody
doesn't want to put a name on to
and so i think it's an interesting
question how does
mathematics and how do formalisms come
into play
with philosophy at all what would any
equation have to say about philosophy
but then once you're in that world of
trying to make models that have some
element of formalism
whether it's natural language or whether
it's another language then
i think you start wandering over towards
the idea of a learning or updating agent
and not needing to take the baggage that
it's a perfect learning agent just that
it is
an adaptive agent of some kind sarah and
then anyone else who raises their hand
yeah this gets for me though um this
gets into a kind of a i don't know the

the clever philosophy word but
um you know this is also why i asked a
question in a prior week about the
markov blanket because it's like
i i picture like a monster truck rally
where two
two fep models are battling it out
basically in my own
brain or whatever and so it's like where
is the
so so in that paradigm where you have
these kinds of objectifications you know
of one fep with another fep like where
the
um then you have questions around like
the boundary between the two
and there's this kind of a there's a
there's a thickness about all of this
that
um kind of gets me in like an infinite
regress
so that's yeah fun alex keefer than
anyone else
yeah so i mean i think this this issue
we're talking about now is really
interesting i
uh yeah i was wondering recently whether
bayesianism is a is a prior
and what that does when you start
thinking about it i think this is
totally
relevant to the the theme of overall
theme of the paper because
well of course it is but i mean it's
directly relevant because um
like it we're talking about things that
get to be so general theories
theories or frameworks or whatever that

are so general that they start to raise
really weird questions but one thing i
was going to say is just that like
i think maybe this is implicit or
explicit what people said already but
like
the fact that we're dealing with
variational inference well for okay
first of all bayesianism even if it's
bayes theorem itself is supposed to like
descend from god or whatever it's like
actually building subjectivity into
into science more right explicitly than
the alternatives i think people mention
contrasting into frequentism but then
we're taking a further step
away from objectivity and saying yeah
and each each creature sort of
learns its own approximation to that um
so i think
um i don't think you can eliminate uh
claims to objectivity entirely um i
don't i
and so this may be maybe this is the the
best we can do
um or at least a step in the right
direction
very interesting point marco then anyone
else
marco then mel yeah i fully agree um so
obviously can't be eradicated and i
think that's kind of the beauty of how
like alex said uh this act of infant
stuff is taking the
the rejection of projectivity even
further it's really
approaching or making at least the the
the

building the foundations for an
objective science of subjectivity
right and and to to add uh to touch back
on the
uh of jokey idea of the bistonian demon
um like alex said right they create
their own particular approximations um
they have to have first instantiated
particular priorities
but more importantly you also noted on
i'm sorry also noted on
infinite regress so so that's another
beauty for me um
there's no problem of if need regress
because traditionally with homunculi and
stuff and controllers or agents or will
theirs is yeah but what is controlling
that and there's this movement
towards smaller and smaller but that's
not the case with
the scenario because the dependency
is not like a vehicle dependent on the
driver
but it's more assistant dependent on
each other and i think that's kind of
the beauty
um where the agents in the perfect demon
scenario
it would be more like um minimal
dependence on the agent's faultiness and
more about how well the niche of the
context
prepares that agent uh for the niches to
be
encountered in their life uh and for
those
you know with attention for buddhism
it's also very beautiful how it relates

to this idea of the mutually arising
or co-origination um it it's just
in essence like alex said it's about
subjectivity
the the the co-dependency of how
densities or how
states or yeah and identities house
densities kind of
adapt each other anyways awesome awesome
mel then tim than anyone else
yeah so i sort of wanna i guess i want
to
push marco maybe to to bring up
a question he raised when you were
giving us the
this sort of like breakdown of this
paper
which i i thought
when he raised i was like wow that's
that's a really difficult question that
i can answer
so um i think that would
push a very interesting discussion
if only because it's a question that i
don't know how to answer and it's a
question of
um it was something about
well aren't aren't physical descriptions
just kind of formal descriptions
at some level anyways so you were you
monica you raised the quest do you
remember do you know what i'm talking
about
no well you raised you raised a question
about about um
about this line that i drew between
a mere you know mathematical formal
statistical description

of something at a a physical description
proper and you sort of
challenge that that such a line exist
which is which is tristan's line
but somehow uh
i i love carl but but um somehow the way
he has always raised that
it feels to me like a cop-out it feels
to me like
some sort of evasion but when you
pushed me on that i was like you know
yeah you're right maybe there isn't
maybe there isn't a hard line
between and and this this sees into
alex's
question as well um
[Music]
it's it's it's all this it's all
continuous this this question of like
well
you know uh
is math natural like is is math an
arbitrary
formal system um
where we've invented all the rules and
it it just has nothing to do with you
know
contingent nature or or is it continuous
you know is it
is it just descriptions of nature at a
very abstract level um
is there a is there a hard line between
information seriously well you know
going full change
is is there uh is there a disappearing
line
between um between
inference techniques between techniques

for bayesian infants
um as a you know just
as a statistical technique and
thermodynamics right you know
statistical mechanics
uh those are the questions that i'm like
oh man i can't i can't that's
that seems to me to be the most pressing
question behind all of this
to me yes math invented or discovered
thermodynamics invented or discovered
fep invented or discovered inference
overall
statistics all these kinds of awesome
interrelated questions
philosophy invented or discovered tim
alex kiefer
then anyone else i was just uh
just kind of the homunculus infinite
regress
thing versus the dependent arising sort
of co-emergence
of the of the i think that contrast is
really
something neat that i don't know how to
explore it but they seem like opposite
ends of
you know two poles of the same kind of
consideration
and i feel uh like i'm repping whitehead
here today but
it also touches on what he talks about
when he talks about the bifurcation of
nature that he decries right so
i mean um uh so when we talk about
taking a perspective and saying that
object
emerges to my perception first of all

because it survived as a self-recreating
thing
over time you know that it uh
you know um and and i did too
we're we're both in the same we're both
the same thing you know what i mean like
it's not like
i'm apart from the world the reason it's
in my world is because i'm in its world
and and the nature's not bifurcated into
the
mind and the physical anyway
awesome point tim because we often say
in active inference that the organism
comes to embody the statistical
regularities of its ecological niche
and so people look out there at the
ecology around them and they can say
okay that's
natural but then the organism is also
natural and wouldn't the generative
model
of its niche be just like its hand be
just another phenotype that's generative
and therefore natural as well so alex
kiefer
then blue then marco
um i i might just i might actually ask
for now i just wanted to drill down more
on this idea of like
mathematics uh what what you know pure
mathematical formalisms and how they
might represent but i think there's a
lot of interesting
stuff on the table right now i'd rather
hear about it at the moment perfect
blue then marco
so maybe it'll i'll wrap it back for

alex because i kind of want to
just touch on this dependent arising
right or like the mutual arising
you know the world exists because it
exists in my mind
um and also the idea of whether or not
mathematics is natural right like is is
this like something that we
invented or something that we discovered
but i would like to make the argument
that that really like
as humans we are also natural right so
so
if it can be discovered in more than one
place at more than one time by more than
one human as many
strong theories in science are they
bubble up together i think that
there is some kind of natural arising or
naturalness
to these kinds of studies
thanks blue marco then anyone else
thanks um so
so uh i think i know what you meant uh
with my comments uh
no i think but but let me allow uh to
push back on myself
um so so i think if we're we we only
have
just descriptions um and the only kind
of extra
qualifiers or properties of these
descriptions as such
that's relevant and grounded is their
etiology or their genealogy where do
they come from
right so if we already talk about
descriptions we already have to be

talking about
systems that can generate descriptions
and so
i don't think there's a hard line uh in
terms of
an essence right that's intrinsic
property but i think the hard line is
more about
their path their genealogy how did that
description came
come to be um and then i think we go
back to
uh language again and i think this is
thanks to um
liam bright shout out to him who
introduced me to karna
and he currently talked about valid
physical languages
um and i think that's kind of the beauty
that the fep
is maybe not perfect for every state of
affair or space of affairs
um but i do think it's true that that
you can transfer a lot of
what we now at least see as valid
physical descriptions to the language
ftp
but more importantly to me the beauty is
the generative aspect so if you
transfer something into um translate
something into active
language of active inference then you
also transfer the generative aspects to
it
so if you transfer them to active
influence and then you explore in the
space that's generated there
and you see a pattern or something then

that hypothesis
obtained or derived in an abstract space
can be also translated back
and so it's the validity of that back
and forth
translational path that i think is more
important
to the question of is it a physical
description right is it is it physically
rooted
rootable if you have these valid
languages i don't know if it makes sense
yeah so small point uh very small points
is
um i've always also taken issue with
carl saying that we embody
statistical regularities about the world
not that it's not true but i think
it's it's it's uh only a partial view
for me what's more relevant is that
there are statistical regularities due
to the fact that the world is systematic
um and these regularities allow this
kind of tethering
right so it's for me it's more about
embodying a bridge with the world
and what you then get at least my
motivation is then that leaves
a more openness to also give room for
the unique
aspect to humanity as in humans are
weird
we're like really weird i don't know why
we're so weird exactly but we're really
weird
and that weirdness has less to do with
cisco regularities and
more about what is evoked or generated

or cultivated
upon embodying this statistical
regulators and upon being faced
with the challenges of dealing with the
world um
and so i guess we have some physical
regulators of the world
and kind of weird irregularities of
ourselves
uh if that makes sense yes yeah we we
swim
in those regularities of the niche you
know if the fish were just the
statistical regularities of water there
wouldn't be a fish there
so it has to be something a little bit
bigger
now another uh cool thing is we we've
been talking about this
uh similarity between the physical
ecology
and then our social ecology which is
also natural so we can think of a
physical example of stigma g
like the ants that are digging out their
nest and they're modifying their
physical environment
which then changes how their physical
behavior is implemented
but with this conversation and in the
literature especially
there's a stigmergy at the level of
information the informational
niche which has different perspectives
within it different agents that have
different access to different kinds of
information
and those kinds of communications in the

informational niche
are natural and the generative models
that we have are natural as well
so there's just so many fun ways that we
can think about all right well
now that we're niche constructing in
this info niche
what kinds of uh things do we want to
put in this
construction that we're working on
together so just like the ant
engineer might say well we could kind of
have a corollary
to this tunnel that branches off this
way maybe there's something useful over
here
oh we falsified that corollary tunnel
it's not useful
it retracts so in the social world in
this informational world
how do we take that natural perspective
and also make it useful for the colony
scott and then anyone else who raised
their hand
just wanted to say with regard to the
fish i think some of you heard the
example i read
research a number of years ago where
they took a semi-rigid piece of plastic
and put it in a stream
and then they varied and increased the
speed of the stream and ultimately the
piece of plastic started undulating with
the periodicity
that resembled the fish and so they what
they realize is the fish doesn't swim
through water it actually passes through
the water

so it was really over deep time fish are
formed by the water
now again not the entirety of the fish
but also formed by other relationships
with other
organisms and media but is similar to
the way that
insects bats and birds independently
developed wings
because they were flying in the medium
of air
but i think i think mel has a point yep
let's go
here yep we'll go mel ben
shannon then marco i
i point there's a it's like a lovely
little video
of a of a dead like salmon in a stream
and it's
it's it's it's behaving
in this way that really seems very
lifelike but sad
um and and i i think i i once
like posted this with the caption like
is this
is this morphological computation and
that made a lot of people there was
there like people were very up in arms
over whether or not that was
morphology
interesting question and synapses and
neurons being physical
what isn't morphological computation
under certain perspectives
shannon then marco than anyone else
that was a brilliant picture of what i'm
thinking in my head right now and
um to figure back on what um both mel

and scott just said
so the dead salmon or the piece of
plastic
you know is passing through the water
in and you can very clearly see how
the environment or the medium that it's
in is forming its behavior
even though it's dead or it's plastic
and
as you get into more weird creatures
creatures with culture
like humans even fish that interact
in groups and communicate with other
fish in a cultural way
then you end up adding different layers
of medium so you don't just have
water you have now the other fish are in
your medium they're part of
this social organism and then with
humans now the other humans are in
are are making another social medium
that you're interacting in
so you have more and more in
the terminology like nested levels of
environments that you're
embodying thanks for that point shannon
marco then scott than anyone else
yeah i'm not sure if it's appreciated
but but i could i'll continue a theme of
linking active influence to some
spirituality slash eastern
traditions it's um so so the notion of
the dao
right which stands for the way i think
it's also
for me it always evokes his resonance
with active inference
um because there's this emphasis on on

on this
the way the tao of how the universe
works efficiently that is unnameable
right and i think that that is also
related to this
issue of reification earlier that there
is
modes there's systematicity there's
rules to how the world works
but fundamentally it's unnameable the
moment you name it it's not the real
way right and uh this kind of finding
the intermediate part
the the one of least effort or as often
as
described no effort away um
this this too is is exactly uh resonant
with that picture of
the life the dead fish in the stream um
so yeah i think that's just in general
there's this theme that i find also very
exciting
how much activation actually helped me
appreciate a lot of
philosophies and tradition not just
eastern but a lot of
let's say forgotten wisdoms and
forgotten insights
from not just religion and not just
spirituality but also arts
right we often see this false dichotomy
of
arts as something frivolous in science
truth
but a lot of insights scientific
insights were actually prefigured
by a lot of artists into conceptual form
and so in my experience active influence

or this generalist
general um lens or perspective allows us
to reappreciate that which has become
too decoupled
from our modern postmodern over rational
of analytic
and a way of engaging with the world
yeah thank you
marco it's almost like dare i say it the
math is not the territory
so we'll go to scott blue sarah and then
alex keefer
so a couple other just observations
along the lines of what
um shannon was alluding to i think in
focal
um philosopher for co not the pendulum
he um talked about pastoral control
which was the idea that the persona of a
person is formed
by their external norms and rules and
laws and
culture that they're exposed to so the
same kind of thing like a plastic toy in
a metal mold
being formed so that same kind of notion
and one of the things i think is
interesting about active inference
for me is that those differentials
what is the motivation for closing
differentials
and the motivation is de-risking in
leverage
grossly stated i think so the idea is if
you have a differential between you and
the environment
you want to protect yourself from the
harm from that differential or exploit

the
opportunities of that differential that
differential is either temperature
differential in the case of a heat
engine
or an information differential in the
case of a von neumann or shannon
extension
of the second law of thermodynamics so
those same differentials you have in
carnot's equation
for heat and cold you need the
differentials and information to have
market exchange
and i think to have beijing exchange
so what was what's the status of a
bayesian process
if you have no differentials between the
input and the expectation
so along those lines what it seems like
is we have these identity stacks this is
what i think shannon was
alluding to not claude shannon but our
shannon here
um and both perhaps but that the
you take these combinations of things
that in the environment that make you
who you are we're like bauer birds we
collect up
these externalities and internalize them
for our expectations
and there's a whole set of them and then
we go out
and identify those that are anomalous
for our expectations so on another call
earlier there's a
someone who talked about the difference
uh narrative and

story and how your internal narrative
and your
story is being told you have to line up
well to the extent that there's a
differential
that motivates activity of interaction
and markets have been described as
places where you find
price discovery and solution discovery
at large scales so the beijing process
seems like a way of incrementally
identifying the externalities that are
anomalous for your expectations
and then being able to internalize them
is kind of that bayesian
process i think so anyway it's something
the
part of the reason that is so appealing
to me is that
i've been arguing for a number of years
that markets and countries and companies
are biological
because their iterations and artifacts
of structures of biological beings which
is us
they happen to take on an abstraction in
an informational
embodiment but they represent those same
processes as biological systems in terms
of those differentials and trying to
render externalities innocuous or
exploit externality
differences anyway there's a lot this is
feeling very
like a there's a strange attractor here
it feels like that's pulling
the conversation in the direction of
that um

generative systems that i define
the last point i define life as being
auto catalytic
entry entropy secreting systems
so these auto catalytic they keep going
and entropy secreting they get rid of
disorder
but the problem is that the neighboring
system has to absorb that disorder
and so how do you manage disorder among
systems
is again when shannon arshan and not
claude shannon
was talking about the different things
the different characteristics
that's the neighbors good neighbors when
they bang into each other in terms of
their
creating anomalies for each other's
expectations
manage the anomalies instead of just
winning over each other so anyway just a
couple of things there
thank you scott blue sarah alex
marco so i'm gonna
skip over what scott just said and go
back to what marco had had previously
said about
um the connections right between active
inference
and eastern religion so i'm a student of
buddhism for many years and i've been
wondering right
do we in fact have free will like if
we're considering
karma and also considering active
inference so this is something that like
i i find this parallel

like karmically we're connected we've
got collective karma with groups with
systems with
you know the universe right in this in
this uh
philosophy so and then we have our own
karma like our previous actions and
and um you know have have collected
collectively
formed us right and so when we're faced
with a choice
like like i wonder do we in fact have
free will because
all we can really do is like if we're
just the product of our karma we only
have one choice
like it's the choice that we're going to
make but obviously that's the only
choice that we could have made
because of all of our previous karma
right so so
and in the same way in active inference
in thinking about active inference
as um like a computational uh from a
computational aspect if this is really
representative of
you know the brain and how how we think
and how you know we operate and how we
make decisions
so thinking about active inference
there's really only one decision that we
could make right like given
the inputs the sensory inputs and like
we have this generative model like given
our model and given the inputs of the
system and our selected policy
there's really only one choice that can
be made like as a result of all this

like collective computation going
forward
anyway it's just something i've been
thinking about for a while blue it's
it's awesome and i just
checked on search engine karma as
multi-scale bayesian prior the title is
not taken yet so anybody who wants to
write that one go for it
sarah alex keeper then marco
yeah just more random into the mix i
guess but um
you know i was i was reading about like
early babylonian
um i don't even know if it was
philosophers but somebody you know said
oh those guys
those guys weren't scientists and
because xyz
and um of course i i disagreed uh but i
was just thinking about just generally
the
um evolution of of science and math
um and there just seems to be this
constant like bifurcation bifurcation
and kind of contrasting this
ever more complex ecosystem to the fact
that we're
um really diminishing our like we're
going through species extinction in the
actual material world
but in the model world we're going
through this this proliferation
and that just like totally trips me out
like what does it all mean
in terms of um connecting those two
things i
i often think about the informational

versus material
and mourn the loss of um
i'm more in the loss of analog actually
like this is a big thing for me and i
don't quite know where to go with it but
um at some point when things became
digitized we really did decouple um
behavior from material in a way that i
think is really significant but i can't
quite get my head around
where it matters and where to where to
cut out the joint i love that phrase
um and how to how to explore that so if
anybody has advice about that also
um but yeah thanks sarah it's like
marshall mcluhan's
work and other media theorists on the
innovation of the written word as one
kind of uncoupling and you're hinting at
even beyond the written word there's a
digital uncoupling and how that
influences
us and especially just you captured it
so well there's a diminishing
biological diversity in some ways as we
modify our environments but at the same
time at least over historical time
periods
we're proliferating there's more songs
there's more scientific theories there's
all these types of things
but what happens if we reduce our
uncertainty about the biological world
till it's just something that can't
support
that kind of theoretical proliferation
alex keeper
then marco then scott oh man um

really interesting stuff going on here
so i mean i originally raised my hand
just
because i wanted to explicitly say that
i do appreciate
marco's um connecting the stuff to
buddhism and things like that um that's
something that like i hope to write
about once i've
established myself as a serious
philosopher a science type person enough
that then i can say things about that
and no one will
react badly um because i think it's
awesome um but
and and this issue you're bringing up
now with like we're kind of enriching
the model space at the expense of the
physical world is really deep and
important i don't know what to say about
it but it's there
um and uh
yeah um i that's uh
the thing that i was that i wanted to
bring up i figured since we're all
talking nicely and getting along about
this these big you know interesting
large-scale themes that maybe um what we
should do is instead
argue about like some detailed annoying
thing
so um so so i just wanted to push um
this idea that so the reason that i
that i don't see the fep as innocent of
representation is just
this might this might be a sort of
because of my provincial view of what
representation is but i figure

if you think of a mathematical model as
like a structure right and then we can
sure you have different windows in that
structure depending on like the
the actual mathematical language you use
to write symbols down
but maybe this presupposes mathematical
realism in some sense but like you've
got a structure
if you've got any system of equations
and then on my view that just represents
whatever
is isomorphic to that structure and it
could be like a
idealized physical system i don't know
you could argue about applications and
where applications come in but
i guess i don't see like how you can
escape
representation um so let's see what
what that does if i say that yep and the
uh
infamous representation wars earlier on
active streams
marco then scott then anyone else who
wants to raise their hand
thanks um uh i'll try to address a lot
of points
and hopefully meaningful um so
i didn't catch everything which you said
scott but but i fully agree that there's
this huge
hugely important issue of dealing with
externalities right so
so i i have some issue with saying that
we simply
throw disorder out in the world i think
that's maybe a consequence of taking the

notion of
exuding entropy a bit too oversimplified
entropy isn't strictly disordered um
because as it's been noted it's all
about actually almost
pretty much yet every information
theoretic measure is
subjective it needs to be relative to
something something often forgotten but
re-emphasize invasionism and now also
machine learning because they're
discovering that it's a huge issue
anyways so the point is that that
um first of all i want to also let you
know
that amongst research and active
influence this is indeed something
that's being
addressed right there's for me one of
the most exciting applications of active
influence
is kind of a philosophy of society or a
philosophy of how we
engage with each other in the world and
how i like to say it and bryce huber
also took that word i think
of co-regulation we regulate each other
so i wouldn't say that that we
intrinsically exude disorder in the
colloquial sense
it's more about the basic knowledge of
active infants you know you perceive
something you do something the question
is to
what extent is that problematic do the
changes that you enact in the world
lead to undesirable consequences that's
the only question

really um but indeed we can't really
address
questions systematically in my opinion
without a framework such as active
influence
and until then we will be myoptic until
then we will take these externalities
for granted
and have to wait we're at the mercy of
our discovery when it becomes too
problematic
such as what we saw with industrial
revolution great great progress in the
terms that you care about
then way later you find out the
consequences the catastrophic
consequences
of externalities at the point than
natural revolution um
ignored um but you know we're working on
it
um and a small side point to add to that
is
um having said that that our exuding of
entropy
is not problematic in itself it comes
also with a re-emphasis
on how the entropy we exude are actually
really great think of children you know
those children are still learning
they've taken all this stuff they don't
understand
and they just express so much chaos
but we love it it stimulates us it
apparently nurtures us you know
people get happy when they see children
playing um
and partly because i think when people

are
doing this chaotic what a chaotic
weird messy behavior in the real world
we're able to kind of understand what's
behind that
and i think that's quite a beautiful way
to look at it because then you can also
see that what maybe messy random
behavior for some person might perceived
as something stimulating
nurturing nourishing uh so so there's a
nice kind of ecology going on there
um and and as for blue about free will
so i i hate the notion free will
mostly used in philosophy because it
seems it almost
always seems like free from determinism
right and i think that's that's that's
already a trap that you cannot fall into
instead it's more about autonomy right
the behavior that you that you engage in
the actions that you
that you that you make to what extent
are they driven by factors that are your
own
factors that are more internal rather
than external
and then of course you would get the
problem of oh well we don't have a self
nothing is true internal
sure but okay then that is simply
because that which is internal
is also cultivated because of the
external but it doesn't take away the
fact i think
that we can say when there's a
generative process to what extent is set
due to idiosyncratic factors

um which i think is beautiful because a
lot of these idiosyncratic factors
are cultivated in part their engagement
with
other people with other cultural
artifacts
and so our cultivation of culture
of narratives of interactions with each
other our practices our rituals
these are all participating to cultivate
this collective
kind of autonomy to what extent can we
as a collective
shape the world niche construct the
world in a way that's more adaptive to
us as an agent
on multiple skills um so
so also i actually don't really like the
notion of district choices but more
fluid navigation imagine if you're
conscious your conscious being
the space that which is conscious you're
more affluently navigating
the density and it expresses itself
maybe as a choice but i think that too
is the real kind of case we talked about
earlier um
so yeah nothing go too long on that um
yeah and i also really like the notion
of uh this uncoupling
so for me what the danger is kind of uh
to use a word used earlier um
there's this body catalytic power to
some
uh narrative to some practices which
shifts
uh the the fitness of these practices to
a more

overly pragmatic mode of behavior one
other way i want to
want to describe this is basically a lot
of our digital technologies create a
training set that is too decoupled from
the actual test set that we will
encounter in the world
um what you see in the kind of rise of
some people might know the notion of
meat space and service space when you
live in service space you're being
attuned to a world that isn't conforming
to
the meat space and i think exactly that
is a danger
and i fully agree with that even though
i'm much very much ancient denizen
i do think this is a problem that hasn't
been fully addressed is dangerous
and the fact that we can't with all
these smart people here that we can't
formulate it properly or characterize
properly
is another indication that this deserves
more attention um
yeah so thanks thanks for all the
thoughts
scott and then anyone else who raises
their hand
so a small fun point and then a big um
fun point so the small fun point is
marco you're talking about children
throwing off wildness and
entropy and that's recreational disorder
right because you know that there's a
thing there it there's two points on
that one one
hot sauce is another one of those right

because it's it's uncomfortable but
people like it right
and so on that there's actually a
physical there was an article in nature
this is a small fun point
article in nature magazine that in your
senses of taste
there are five senses right there's
actually two channels good and bad
the good channel is sweet which is
carbohydrates and umami which is
proteins
the bad channel is bitter and sour which
are poisons
typically an unripe fruit and so salt
doesn't have its own channel
salt goes from the good channel until
the level of blood salinity and then
switches to the bad channel
so the reason for mentioning that is in
food you use bitter and sour
as highlights as recreational disorder
right and so they're and they're because
they're not indicative usually of
consumables
in nature so that was just an
interesting side light on that
it's a similar to this notion in the
dentist there's this research now
that if you have a very painful dental
procedure you should actually finish up
the procedure with a little
extra pain but not but a soreness type
pain not a difficult pain because the
mind remembers the last pain experienced
so the the article said do you want a
little pain with that
and the idea was extra irritant and the

ada said they can't
because of the do no harm they can't
administer extra pain
as a matter of policy but the idea was
to hijack the sensation the memory of
the bad experience with the lesser pain
which is kind of interesting so that was
the small fun points
let me get to the bigger point which i
think um is
uh it goes back to what sarah was
alluding to
so this is that idea of symbolic and
physical beings
okay this is something i've been playing
with for a little while it takes a short
period to describe
and so let me try that so what if put
put aside all your notions of causation
for a minute i'm going to go back to
something that seth lloyd raised for me
a guy at mit wrote programming the
universe and he asserted in his book
that all interactions since the big bang
have increased exponentially
all interactions in the universe okay so
two two particles hit each other then
they get four particles it's 16
particles whatever
and then that iterates out to us okay so
moore's law
you have this exponential increase in
transistor density on chips
that has led the fifth order effect of
that to an exponential increase in human
interactions okay
because the chips are more ubiquitous
they can digitize more interactions

we're having this discussion now is not
possible before
okay so the more interactions are
happening exponentially let's go with
that
for a second suspend disbelief and just
imagine that all interactions are
increasing exponentially
okay what happened with humans is
physical space did not offer enough
vectors of engagement
to contain interactions increasing
exponentially and so we had to go to
symbolic space
so there were a couple a couple of other
i'll get back to that in a second
so think of clonal reproduction versus
um
sexual reproduction clonal reproduction
you resemble your parents
and for niche space and again this is
not intelligent design but
stay with me for niche space clones are
not sufficiently varied
so that if the niche space is changing
rapidly the clones won't have as much
survivability mutation random mutations
can introduce that noise that can allow
the accommodation of
that change in physical space i think
what's happening with humans
is that we are becoming information
beings increasingly
and i say that to people like yeah yeah
yeah and then i say hey
you know that 401k that you have if
anybody has any retirement savings
the i say to them do you think that's a

pile of groceries
or or a house waiting for you to retire
that is data
that is nothing the system goes you got
nothing it's
totally symbolic so one of the if and to
the extent
that interactions have and their
artifacts are increasing exponentially
throughout
time throughout deep time it was
inevitable that we moved to
information space because the
the phase space available for solutions
to our problems was
insufficient in physical space we needed
more dimensionality
in order to address things and that's
what language does for us it opens up
dimensions it opens up abstractions it
opens up things called war
it opens up things called stock markets
it opens up things called
insurance law governance
all these things that allow for us to be
human and the thing that separates us
from other organisms to some extent is
that language is where the mind exists
because that's the symbolic element of
who we've become as information beings
there's a lot there
but i think one of the things that's
been i've been playing with just try to
understand what this might be
is what does it mean to be converted
increasingly to information beings
and not we're not still have a physical
presence but

this came up first and last point here
this came up first in the identity work
i did years ago
where people are saying oh you can't
control me on the internet i said sure
you can
a government a sovereign can put you to
death they have the monopoly legitimized
violence
or they can silence you which is we've
seen happen now and so the physical body
if i die right now while i'm talking
which some people might like with this
long thing i'm
saying here if i died right now
immediately
my legal agency to do things on the
internet would change
and it's an entanglement it's almost
like a quantum entanglement because it
would happen instantaneously i cannot
have agency if i'm dead
so anything pre-programmed in my
phys in my symbolic self could keep
running but the
physical world then doesn't respect you
as
having a will anymore
so there is an entanglement so as long
as governments continue to control the
physical body
and have the monopoly of legitimate
violence they can control the digital
body because putting you to death
de-animates the in
and we have uh dave then
sarah then stephen and we'll steer it
back to the paper

and also get towards the last thoughts
that each person
wants to share so again dave sarah
steven and then
anyone else
okay the um when the kids make a lot of
chaos
uh you look at the the
especially one-on-one games that kids do
they are largely dominance games
one kid is going to tend to monopolize
being the
the dominant figure the cowboy the
sheriff
for a while and then switch back now if
they hang with that
role for more uh more than about 60
percent
of the time eventually it gets actually
almost sadistic
and it becomes too aggressive the little
and
things get out of hand the dominated kid
cries
runs off the smart mommy just kind of
gives a little pat
and lets the kids go on their way the
overprotective mom though
almost takes over the bullying and makes
it a terrible thing
which means that kids can't go back and
achieve restoration so when the
when the the child who has been
paranoid doesn't have the reassurance
that oh well okay the world does
eventually become okay
and the kid who's been in the sadistic
infantile mode

doesn't get the reassurance well even if
i was evil mommy for a while
i'm not evil mommy anymore now i'm good
mommy
so another reason to let your kids whack
on each other and get dirty
okay thank you sarah stephen and then
anyone else
oh um scott's comment kind of reminded
me of what i
what i unconsciously was bridging um
with respect to
information scott said something i
verbatim i don't remember but basically
that um something about language and
for me you know language uh languages
as with ecosystems as with um species
extinction like languages
are going extinct and so languages are a
really interesting bridge to this
you know information space um and it
seems that
that fact of language is going extinct
seems to contradict something you said
scott but
again because this is i feel like mel's
show i want mel to like
hook this into whatever she's working on
if if
that makes sense for her steven
tim then mel or anyone else
stephen muted
all right sorry sorry i'm really i'm
just trying i was looking for a paper
that i was trying to find on something
this okay um
so what i was gonna say is that
there's there's a lot of stuff coming up

here absolutely
um and one thing i like about this paper
is it
it sort of starts from the free energy
principle and sort of really
pays attention to that and the active
inference and i think that there is
something important
like because we get into this
conversations about information and
stuff is
it's like the free energy principle
there's a collary for active inference
of these stages
and different scales and then once that
scaling is happening there's also
a free energy principle that sort of
scales up
anyway the natural nature of how these
these societies try to work with these
models
and there's a paper that was i think in
the physics review of life
and i'm trying to find it but it sort of
talks about how we
naturally try and reduce free energy
over time and i think this is
particularly what's happened
since capitalism in the last 500 years
or
200 years particularly we want to reduce
or there's a tendency in different ways
to try and reduce
the amount of free energy out there and
we can do it more and more because
we can construct and our focus is
constructing
um these um larger scale

um societies social networks ecosystems
a super organism
as uh mel talks about it and we're kind
of
we're sort of constructing this niche to
such an extent that we just take that
for granted
and these perspectives on things and the
rarification is almost
um de facto but there's also
probably more traditionally in history
and it's probably also true if you think
like what do you do when you're out in
the middle of the ocean
you can't control the ocean so now you
have to try and adapt whatever the winds
are doing
based on what you've got so this
the organism like as an active inference
process
to try and minimize free energy and just
to stay
attuned and alive in a niche
um is is something that the sort of very
foundational
understanding from physics and chemistry
sort of
still alludes to you know and then we
have these higher the
this ability to construct through action
and um and this is where
we can start to feel that these actions
that we do
like we talk about that we are
information but
this is this is the danger of the
verification of information because
we at some extent because we've built on

top of things yeah we kind of construct
things in our niche
which makes it and it is kind of an
informational
hole in our niche but we're still a
being
that has to um
dynamically engage with that niche
and that information doesn't travel
like unlike in traditional inactivism
that information doesn't come in as like
that's a tree it comes in again in
amongst all this noisy
um statistical variations
and we infer from that so there's this
question about
that danger the information starts to
get rarified and is
and becomes seen as something being
transmitted down the channel
but that's only something that
effectively emerges
once some processing is done so i think
i think that's that's what sort of came
up for me thank you steven
so in our last couple of minutes we will
go to
tim marco and then anyone else so let's
just keep it brief and summary and
remember that next week in 14.2
we're going to be on the same paper so
no need to cram it in in the last minute
maybe raise some questions or ask
somebody about their perspective
so that we can mull on it in the next
week and then come back next week with
the links ready and with the ideas ready
with the questions ready

and read the paper another time so tim
marco and then anyone else
yeah just uh briefly uh the child's play
maybe it's obvious and it's like what
everybody already knows about
things but uh it struck me when you were
talking about it that that's kind of
like
active inference on hyperdrive right
because the the child's play is this
you know generating these fanciful
hypothesis you know like the imagination
is just going and then that puts them in
a position to enact with interact with
the world
and maximize surprisal as they run all
over the place and
trip over things right so it's like it's
learning on
hyperdrive and then the only other thing
i want to say is wrapping up was as i
was going through and reading the paper
and all those
attempts to answer the question what is
the free energy principle in terms of a
category
there was one kind of startling
reductive idea that came to mind
it seemed really silly but i'll throw it
out there it could it just be like an
idealization like frictionless
you know because um it's kind of it's
compared to the um
principle of at least action right but
but if you look at the actual free
energy principle
through the layers of statistical
mechanics like all those layers

going into that uh kind of translation
uh the free energy principle almost just
seems like since we can't really see all
that
through all that cloudiness of
statistical and probability and such
and um it seems like just an
idealization
a way to say we can do something with
this even though it's this distant thing
that's gone through this probability
machine
but we can do things with it because we
have this idealization that we call the
free energy principle so it's just
another another option for your what it
is thing maybe it seems silly i don't
know
yep mel then marco yeah so
so interestingly um we've had
we've had in philosophy of science we've
had series of how
science works for a long time and we've
gone through many kinds of
uh popular accounts and mainstream
accounts of how science works
and um these have traditionally
emphasized um
series um um and accuracy
right you want you want you want a an
accurate true representation of of
what's going on in nature right um
but uh and and false circulation this
sort of thing um
the the kind of and rogue view right now
the the the idea du jour
is is really in a lot of ways focusing
on models focusing on

idealization focusing on pluralism um
there are a lot of great books on this
um michael weisberg's simulation and
similarity
fantastic book great introduction to
models
some of the big debates models um steven
downes has a new like
tiny little just introduction that just
gives you a breakdown of all the
literature and models
and um angela pertochnick
in my department has a fantastic book
called idealization in the aims of
science
and then um so what was funny to me is
like i wrote this
paper i wrote this paper and then
the the math is not the territory and
then a few weeks later someone
um uh a professor of my apartment clean
line department
announced that they were running a
course for the spring term um
on on this new book by uh
called when maps become the world
which is actually a fantastic book i'm
really enjoying this book
and it's touching on all of these things
that you guys have been
talking about like kind of different
modes of representing and
do these things represents or not um and
how
the kind of feedback between our maps of
the world
and like cultural differences and how we
how we

represent things and um
where the world is going these kinds of
kind of big picture questions and
it's a really lovely read um
the idea with with models right
is that what they do is chiefly
idealized
right um if if you look at like the the
the whim facts
um the one side account of of what's
going on in modeling
right the idea is that um
it's it's not meant to be
a maximally accurate depiction of some
natural system or some natural process
um it's not it's fidelity
to the world that makes the model useful
but in fact it's the idealization
it's what gets coarse-grained black box
away you know it's it's it's it's about
finding is but latching on to some kind
of key feature
of the system of interest and that's
always mediated by kind of what our
what our questions are as researchers
what our interests are in these natural
systems and researches so it's kind of
value-laden
in a way um and it's discarding the rest
it's discarding the information that is
inessential to
the question at hand so so idealization
is at the core of what a model is
awesome response so any very
rapid final words this is the true
lightning round marco
scott alex kiefer oh dear i'm not good
at lighting around

um uh well okay very brief
so so i agree about the importance of a
notion of friction but i do think it's
already
uh you get it for free my colleagues say
you get it for free with active
influence i think because
you have two situations where i think
you will get friction
one is pathologically stuck when you're
overly rigid
because there's no alternatives of
sufficient credence the base factor is
too high
so you can't move out of it right so
instead of saying there's a present
friction
it's more like there's an absence of a
lubricant or something or an absence of
a passive
and the other side if you have too many
possibilities your base factor can't get
too high because everything's kind of
similar
so you get stuck in that and i think for
me those are two forms of friction slash
inertia um i just very briefly want to
wrap up
that big theme of how we scaffold all
these things upon society which is
attractive upon extractions
and the many many dangers that lie there
where
indeed you would have i think an auto
catalytic
pitfall of um trying to cope with the
complexity of the world but falling into
heuristics

and when these heuristics are used to
minimize your
experience consequences of excessive
free energy
and then adopt that heuristic then it
will spread this kind of self-selecting
manner
as in the people exposed to these
notions such as for example q anonymic
conspiracies
they heuristically connect everything
together and so you seemingly experience
coherent as long as you don't stick too
far or commit
to proper epistemic virtues and and
indeed this will become worse and worse
unless we have
an alternative a better ground at least
in my opinion
um and i do want to push back a little
bit on this
uh mostly true claim that the models
are just vitalization but i think what's
really really interesting is not just
models as presented in papers models as
defined in maths
but take into account the human
enterprise that is science
because everyone has to adopt this model
not just program it didn't some code
way more interesting is what happens
when you adopt or incorporate or
internalize
an idealized model and for me that's the
expressiveness vacuum
upon incorporating it you will
de-idealize it you will
fill it up you will make it your own

particular and that expressiveness my
opinion is what makes
it interesting and what mel has really
laid the groundwork for it might be
but yeah i wish we could have talked
more about the paper thanks marco
again it's dot two it's not about
answering every question in the dot one
or even the dot two it's about staying
excited
about including as many people in the
conversation hearing from everybody's
perspective if you're curious and
listening
joining the key base actinflab.public is
a great way to get engaged with action
not just inference so again single kind
of closing thoughts but less personal
review paper my ted talk but more
what are we going to move forward to
into 14.2 not that anyone did that
scott alex kiefer steven sled
the painter kandinsky said that
violent societies yield abstract art
and one of the things i always wondered
is is the reverse also true
is abstraction a form of violence
and so that's something i i posit to you
that 2008 financial break was
abstraction
of mortgage instruments that caused a
violence
of people who were kicked out of their
homes the math
so the math of the mortgage was not the
territory of the lived experience and
the need for housing
and so that is where the disconnect

happened so something to be aware of in
terms of abstraction and the possibility
of it being a form of violence
and the monopoly of legitimate violence
being a definition of the sovereign from
mills you might think about the monopoly
of legitimated abstraction
being a form of sovereignty thanks thank
you
scott alex keefer stephen and then sarah
um so one thing i'm still interested in
talking about
um potentially next week is i guess this
probably is like a microcosm of some of
the stuff we've talked about but um
sort of the jaynesian perspective in
thermodynamics and
in physics and how this relates to all
this um because
um i think there's potentially something
interesting so i
one thing i like about this paper i will
be brief here one thing i like is that
it goes through sort of the history of
the formalism of the fep and where it
came from
and i think in part that's meant to cast
doubt on
any direct connection between the fep
and physics but at the same time
the jnz perspective in physics is
something the fep inherits from and it
sort of already maybe gets epistemic
stuff into physics at a lower low level
so i want to talk about the relationship
between
um entropy and thermodynamic sense and
entropy in the shannon sense

if we have a chance thank you we're
gonna go to mel um with uh first author
privilege and then again anyone can drop
off if they need to go but let's
bring it to a close together otherwise
mel stephen sarah
yeah actually um i i would really love
to talk about
that point that you raised um and
my other reviewer i know who it is
they find it um my other reviewer uh
andrew corcoran actually a great guy um
raised that there's this new paper um
like gothwald and brown published
december 3rd
2020 that is in
plus computational biology um
and it's uh it's it's really getting
into changes so
it's really getting into free energy
it's really getting into the jameson
perspective it's like the first
it's like the first time
anyone has really delved into
james and the fep um so
i'd love to discuss that because i
haven't i haven't read it
um yet and i'd love to discuss that
thank you now julian james which jeans
is that
no et james so so max thank you maximum
entropy principle i can i can drop this
link in the in the comments below and
i'd also like to talk about
representation
like different different approaches
representation and how models represent
if they do necessarily excellent steven

and i'll go back to just my regular
camera steven go ahead and then anyone
else
yeah i think the i would like to hear
more on that uh james the entropy the
maximum entry
approach because i think the whole
entropy
agonic thing is really interesting i
think that's this paper like i say is
really tapping into that
and uh and i suppose one thing i'd like
to say is we talk about models
is um we're using models to get the
dynamics
and there's a slight trap because well
that makes sense
um but you have to take a perspective
and verify it and it's like this modern
world we just
do it but the the key thing that's also
present with the whole
flows of information is and is that your
models can reveal
other ways of knowing which are not
perspectival
and are not themselves so the
interesting thing
about dropping something out of the
active inference
model isn't the model is the dynamic
shifts that the graphs show
when the model's running so that and
that can tap into indigenous ways of
knowing because the thing with
indigenous people i'm saying earlier is
that they're trying to attune to the
world

not say what the world is and model it
and that maybe is what we need to do
more
to actually survive on this planet as
it's an ecological collapse because we
might need to
sense what is it that we need to
dynamically
embody that isn't entirely more
liberal so you know we could talk about
that that's enough for me yes
sarah did you have something okay
marco any last thought
i just want to um echo what's been said
by mel and
alex i also think that one of the most
interesting things especially for the
next session
to talk more about the relation with
thermodynamic and information theoretic
entropy
especially because there's one sentence
in males that i just want to point out
so it says the elements of the framework
do not map onto any known features of
real-world systems
at least not with any more granularity
or specificity
than the constant dynamics of such
systems but i found a very short
sentence because it seems that
the granularity or specificity of their
causal dynamics would be
sufficient to act as a model right and i
think the really really interesting
thing that hasn't been exported enough
but
but mel griff's a good uh starting

ground for is
asking more about the relation between
the topic
thermodynamic and information theoretic
entropy where i think we should
maybe move more or emphasize the the
perspective of interpreting entropy as
distribution how fast homogeneous how
distributed is it
as a tendency to kind of flat out to
spread out
in the thermodynamic sense and then i
think with that conceptual metaphor
it's not really a metaphor it's just
true but with that concept in mind i
think it's much
easier to have a conversation about the
relationship between information
theoretic
and thermodynamics that being said
looking forward to uh
thanks so much mel any last thoughts as
the
first and last author otherwise i'll
close it out
phenomenal thank you this is great yep
it's great
mel thanks so much for not just the
paper but for engaging with us you know
asynchronously on twitter and
synchronously
on video chat all of the participants
there was so much we brought up today
and it was just awesome to hear what
everyone
had to share so if you're listening live
or in replay
you're part of the community of practice

and we want to just
have you participate however so if
you're curious about something
post a question if you want to know how
you can participate just
reach out to us through any number of
these mechanisms
other than that on january 26th we will
be right back here
for 14.2 and we'll deal with some of
these questions
everybody will come with a few new
digested thoughts
a couple of polished questions and we
will go from there so thanks again
everybody
and we'll see you next week